

RECEIVING REPORT

Receiver No.: 18424655

Document Date: 03/27/2025

P.O.#: 5300130329

Posting Date: 03/27/2025

Item #: 1

Expiration Date:

Batch#: ACE0121235

Insp. Lot: 000001298048

Plant: ACE

Plant Description: Clearfield-Solumina

Vendor Name: HEXCEL CORP

Vendor Addr: 1913 N KING ST

SEGUIN

TX 78155

Quality Inspection

Unloading Point:

Vendor No.: 90069901

Buyer Name: Keith Larsen

SQAR Code:

Project ID: ASBUX

Test Lab:

WBS:

Inspection Lot/Type:

Rev No.: 01

Material: MSRR4040-VAFR

Received UOM: LB

Base UOM: LB

Base UOM Received Qty: 1550.020

Vendor Mtl. Number:

Mtl.Description: SATIN WEAVE CARBON FIBRE REINFORCED BISM

Mtl.Specification:

Alt. Material #:

Matl. Basic Data Text:

MSRR4040-VAFR:SATIN WEAVE CARBON FIBRE REINFORCED BISM

Bill of Lading: 92SPS055277

Delivery Note: 92SPS055277

Receipt Created by: N15071

Received Qty: 1550.020

Head text:

Receipt Delivered To: _____

WAREHOUSE INFORMATION



Plt: ACE



Part Number: MSRR4040-VAFR



Qty: 1550.02



Receiver No: 18424655

Storage Bin:



Order Number:

Warehouse: LIN1



Stock Type: Q



WBS Element: A-46Z01-NW.N46Z01000713

Inventory WBS: Y-ROLLS

Bin:

Storage Location: ACED

Item Text:

SATIN WEAVE CARBON FIBRE REINFORCED BISMALEIMIDE PREIMPREGN
ATTENTION: The revision level for this part is: 4
NG PROJECT: 46001.IN.INV.0RAW, PRIME CONTRACT:
Ship via: Best Way.

Qty. Insp.	Insp. Stamp No.	By Inspector	Qty. Rej.
Rej. No.	Stocked By	Date Stkd.	Stk. Loc.



26 March 2025

CERTIFICATE OF ANALYSIS

Customer: Northrop Grumman Systems Corporation

Purchase Order No: 5300130329 Part Number: MSRR4040-VAFR

Material: F3C 433/M65: 38%; 285AW; 50"

Specification: MSRR4040, Issue 4, dated November 2012.

Certificate of Conformance: Hexcel lot 92M0103861 has been manufactured and tested in conformance with specification MSRR4040, Issue 4, dated November 2012.

Lot No: 92M0103861 Date of Manufacture: 17 February 2025

Storage Life: 12 months from date of manufacture when stored at 0°.

Resin Lot No: 92M0103522 Manufactured by: Hexcel

Glass Lot No: 31W0026323 Manufactured by: Hexcel

I. Prepreg Physical, Chemical and Mechanical Properties
See Certificate of Analysis Report.

II. Individual Spool Physical Properties
See In Process Control Acceptance Testing Data.

Note: See Packing List for date of shipment, actual quantity and spools shipped.

Laney Defa 



Matrix Testing Laboratory
5400 S. 6700 W
West Valley City, Utah
84118

Certificate of Analysis Report

Specification: MSRR4040 Rev. Issue 4
Configuration Number: CN1301
Material Type: F3C 433/M65;38%;285AW;50"
Lot Number: 92M0103861
Order Number: 92M0103861

Report Printed: 04-Mar-2025 12:07:04PM
Status: **PASS**
Approval: **CARTERKC**

1. Tensile, SBS, and TG Panel

Dimensional Fiber Volume

Roll/Panel	Test/Result:	Requirements:	Values:
92M0103861001/001	Laminate Density By Immersion Avg (g/cc)		1.5086
	Laminate Density By Immersion Std (g/cc)		0.0028
	Laminate Density By Immersion CoV		0.18
	Dimensional Fiber Volume (%)		53.49 - 53.00 - 52.43
	Dimensional Fiber Volume Avg (%)	50 - 56	53
	Dimensional Fiber Volume Std (%)		0.531
	Dimensional Fiber Volume CoV		1.00
	Void Volume (%)		1.0 - 1.0 - 1.0
	Void Volume Avg (%)	MAX= 2	1
	Void Volume Std (%)		0.017
92M0103861018/002	Laminate Density By Immersion Avg (g/cc)		1.5105
	Laminate Density By Immersion Std (g/cc)		0.0033
	Laminate Density By Immersion CoV		0.22
	Dimensional Fiber Volume (%)		54.51 - 53.86 - 53.43
	Dimensional Fiber Volume Avg (%)	50 - 56	54
	Dimensional Fiber Volume Std (%)		0.544
	Dimensional Fiber Volume CoV		1.01
	Void Volume (%)		1.2 - 1.2 - 1.3
	Void Volume Avg (%)	MAX= 2	1
	Void Volume Std (%)		0.040
	Void Volume CoV		3.32

Short Beam Shear Stress-Warp-Ambient

Roll/Panel	Test/Result:	Requirements:	Values:
92M0103861001/001	SBS Strength (MPa)		71.6 - 72.7 - 75.8 - 81.2 - 78.4
	SBS Strength Avg (MPa)	MIN= 65	76
	SBS Strength Std (MPa)		3.97
	SBS Strength CoV		5.22
92M0103861018/002	SBS Strength (MPa)		73.8 - 79.9 - 74.8 - 83.2 - 79.3
	SBS Strength Avg (MPa)	MIN= 65	78
	SBS Strength Std (MPa)		3.89
	SBS Strength CoV		4.97



Matrix Testing Laboratory
5400 S. 6700 W
West Valley City, Utah
84118

Certificate of Analysis Report

Specification: MSRR4040 Rev. Issue 4
Configuration Number: CN1301
Material Type: F3C 433/M65;38%;285AW;50"
Lot Number: 92M0103861
Order Number: 92M0103861

Report Printed: 04-Mar-2025 12:07:04PM

Status: PASS
Approval: CARTERKC

Short Beam Shear Stress-Warp-150C(302F)

Roll/Panel	Test/Result:	Requirements:	Values:
92M0103861001/001	SBS Strength (MPa)		66.2 - 68.5 - 66.4 - 67.9 - 65.6
	SBS Strength Avg (MPa)	MIN= 50	67
	SBS Strength Std (MPa)		1.25
	SBS Strength CoV		1.87
92M0103861018/002	SBS Strength (MPa)		64.3 - 65.3 - 64.5 - 66.2 - 64.9
	SBS Strength Avg (MPa)	MIN= 50	65
	SBS Strength Std (MPa)		0.73
	SBS Strength CoV		1.12

Tg By DMA

Roll/Panel	Test/Result:	Requirements:	Values:
92M0103861001/001	Tg by DMA in °C (° C)		266.4
	Tg by DMA in °C Avg (° C)	MIN= 225.0	266.4
92M0103861018/002	Tg by DMA in °C (° C)		262.3
	Tg by DMA in °C Avg (° C)	MIN= 225.0	262.3

Calculated Thickness/Ply

Roll/Panel	Test/Result:	Requirements:	Values:
92M0103861001/001	Calculated Thickness/Ply Avg (mm)	0.28 - 0.32	0.30
	Calculated Thickness/Ply StDev (mm)		0.0051
	Calculated Thickness/Ply CoV		1.50
92M0103861018/002	Calculated Thickness/Ply Avg (mm)	0.28 - 0.32	0.30
	Calculated Thickness/Ply StDev (mm)		0.0025
	Calculated Thickness/Ply CoV		1.09

Tensile Stress, Modulus, and Elongation-Warp-Ambient

Roll/Panel	Test/Result:	Requirements:	Values:
92M0103861001/001	Norm Tensile Strength (MPa)		982.3 - 970.9 - 978.6 - 959.2 - 959.3
	Norm Tensile Strength Avg (MPa)	MIN= 700	970
	Norm Tensile Strength Std (MPa)		10.68
	Norm Tensile Strength CoV		1.10
	Norm Modulus (GPa)		63.5 - 62.2 - 64.0 - 61.9 - 62.5
	Norm Modulus Avg (GPa)	MIN= 55	63
	Norm Modulus Std (GPa)		0.896

Certificate of Analysis Report

Specification: MSRR4040 Rev. Issue 4
Configuration Number: CN1301
Material Type: F3C 433/M65;38%;285AW;50"
Lot Number: 92M0103861
Order Number: 92M0103861

Report Printed: 04-Mar-2025 12:07:04PM
Status: **PASS**
Approval: **CARTERKC**

Tensile Stress, Modulus, and Elongation-Warp-Ambient (Continued)

Roll/Panel	Test/Result:	Requirements:	Values:
92M0103861001/001	Norm Modulus CoV		1.43
92M0103861018/002	Norm Tensile Strength (MPa)		979.3 - 961.1 - 993.6 - 1005.8 - 989.6
	Norm Tensile Strength Avg (MPa)	MIN= 700	986
	Norm Tensile Strength Std (MPa)		16.78
	Norm Tensile Strength CoV		1.70
	Norm Modulus (GPa)		63.5 - 62.2 - 62.1 - 63.4 - 62.2
	Norm Modulus Avg (GPa)	MIN= 55	63
	Norm Modulus Std (GPa)		0.717
	Norm Modulus CoV		1.14

2. PHYSICAL PROPERTIES

Volatile content @100°C in pan

Roll	Test/Result:	Requirements:	Values:
92M0103861001	Volatile content @100°C in a Pan (%)		0.6 - 0.5
	Volatile content @100°C in a Pan Avg (%)	MAX= 1	1
92M0103861018	Volatile content @100°C in a Pan (%)		0.5 - 0.5
	Volatile content @100°C in a Pan Avg (%)	MAX= 1	0

3. PHYSICAL TACK

Tack

Roll	Test/Result:	Requirements:	Values:
92M0103861001	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30
92M0103861004	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30
92M0103861007	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30
92M0103861008	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30
92M0103861010	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30

Certificate of Analysis Report

Specification: **MSRR4040 Rev. Issue 4**
 Configuration Number: **CN1301**
 Material Type: **F3C 433/M65;38%;285AW;50"**
 Lot Number: **92M0103861**
 Order Number: **92M0103861**

Report Printed: 04-Mar-2025 12:07:04PM

Status: **PASS**
 Approval: **CARTERKC**

Tack (Continued)

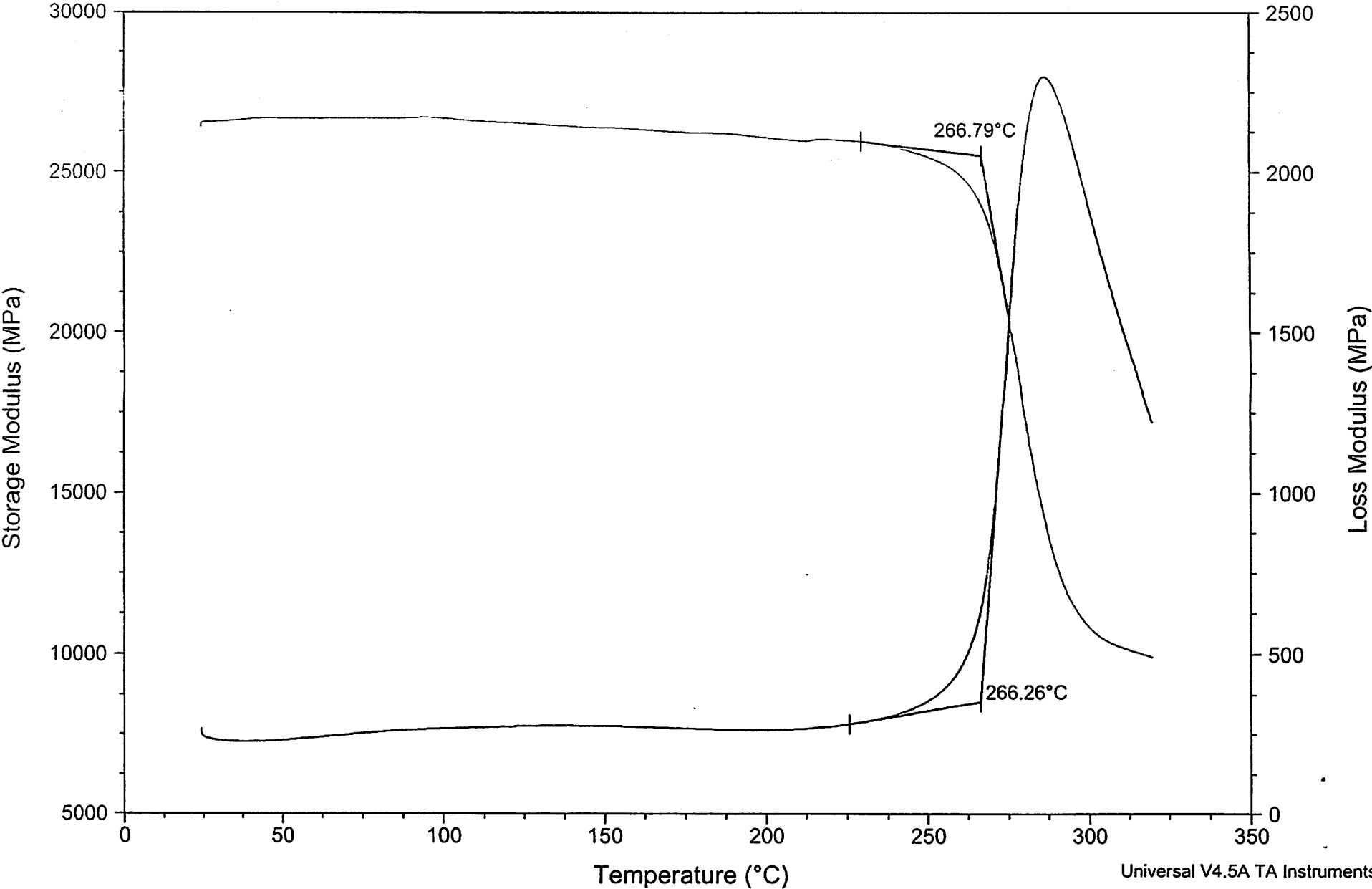
Roll	Test/Result:	Requirements:	Values:
92M0103861012	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30
92M0103861014	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30
92M0103861016	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30
92M0103861018	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30

End of report

Sample: M65 92M0103861-001 SPL#001
Size: 35.2300 x 9.9200 x 2.4200 mm
Method: Temperature Ramp

DMA

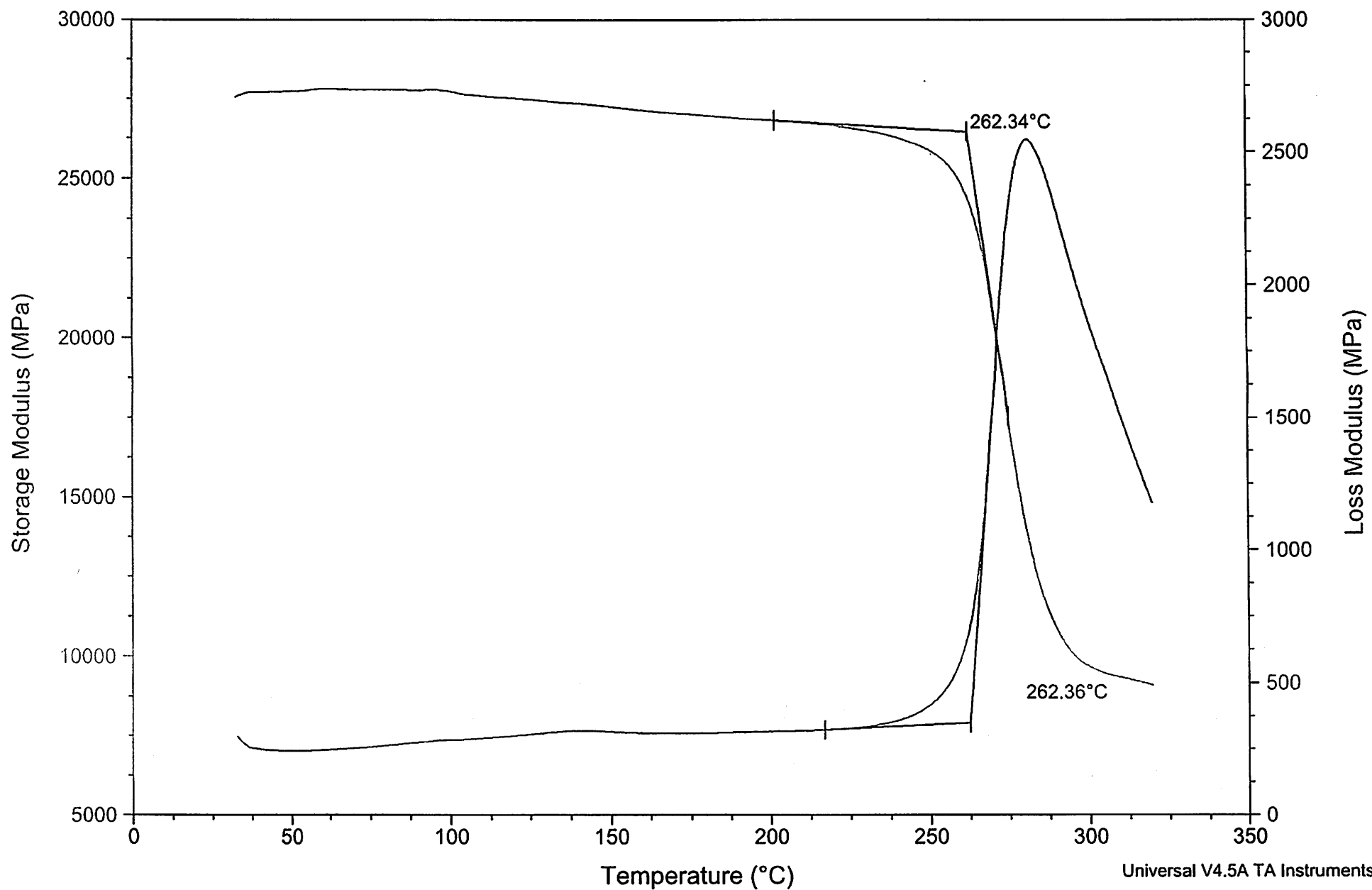
File: \\...\\M65 92M0103861-001 SPL#001.001
Operator: EG
Run Date: 02-Mar-2025 12:37
Instrument: DMA Q800 V21.3 Build 96



Sample: M65 92M0103861-002 SPL#018
Size: 35.2000 x 9.9400 x 2.4000 mm
Method: Temperature Ramp

DMA

File: \\...\\M65 92M0103861-002 SPL#018.001 .
Operator: EG
Run Date: 02-Mar-2025 14:05
Instrument: DMA Q800 V21.3 Build 96



IN PROCESS CONTROL

ACCEPTANCE TESTING DATA

RC / AWT Report for lot: 92M0103861

3/26/2025

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Spool	Re - test	Rep	Init - Weight	Final Weight	RC	AWT	PAW	Width	Stat.	Tech	Date
92M 0103861001	0	1	4.5808	2.7907	39.1	279	458.1			SAVASBX	02/17/2025
		2	4.5877	2.7827	39.3	278	458.8			SAVASBX	02/17/2025
		3	4.5707	2.7765	39.3	278	457.1			SAVASBX	02/17/2025
		Avg.	4.5797		39	278	458	50.25	A	SAVASBX	02/17/2025
92M 0103861002	0	1	4.6164	2.8105	39.1	281	461.6			SAVASBX	02/17/2025
		2	4.6111	2.8087	39.1	281	461.1			SAVASBX	02/17/2025
		3	4.6189	2.7894	39.6	279	461.9			SAVASBX	02/17/2025
		Avg.	4.6155		39	280	462	50.25	A	SAVASBX	02/17/2025
92M 0103861003	0	1	4.5976	2.7619	39.9	276	459.8			SAVASBX	02/17/2025
		2	4.5594	2.7859	38.9	279	455.9			SAVASBX	02/17/2025
		3	4.5552	2.7771	39.0	278	455.5			SAVASBX	02/17/2025
		Avg.	4.5707		39	278	457	50.25	A	SAVASBX	02/17/2025
92M 0103861004	0	1	4.5607	2.7984	38.6	280	456.1			SAVASBX	02/17/2025
		2	4.5994	2.7804	39.5	278	459.9			SAVASBX	02/17/2025
		3	4.5423	2.7959	38.4	280	454.2			SAVASBX	02/17/2025
		Avg.	4.5675		39	279	457	50.25	A	SAVASBX	02/17/2025
92M 0103861005	0	1	4.5987	2.8127	38.8	281	459.9			SAVASBX	02/17/2025
		2	4.6056	2.7765	39.7	278	460.6			SAVASBX	02/17/2025
		3	4.5797	2.8179	38.5	282	458.0			SAVASBX	02/17/2025
		Avg.	4.5947		39	280	460	50.30	A	SAVASBX	02/17/2025
92M 0103861006	0	1	4.5418	2.7786	38.8	278	454.2			SAVASBX	02/17/2025
		2	4.5864	2.7859	39.3	279	458.6			SAVASBX	02/17/2025
		3	4.5895	2.7761	39.5	278	459.0			SAVASBX	02/17/2025
		Avg.	4.5726		39	278	457	50.30	A	SAVASBX	02/17/2025
92M 0103861007	0	1	4.6147	2.7931	39.5	279	461.5			SAVASBX	02/17/2025
		2	4.6428	2.8115	39.4	281	464.3			SAVASBX	02/17/2025
		3	4.6311	2.8050	39.4	280	463.1			SAVASBX	02/17/2025
		Avg.	4.6295		39	280	463	50.30	A	SAVASBX	02/17/2025
92M 0103861008	0	1	4.6117	2.8349	38.5	284	461.2			SAVASBX	02/17/2025
		2	4.6362	2.8142	39.3	281	463.6			SAVASBX	02/17/2025
		3	4.6015	2.8216	38.7	282	460.2			SAVASBX	02/17/2025
		Avg.	4.6165		39	282	462	50.25	A	SAVASBX	02/17/2025
92M 0103861009	0	1	4.6189	2.8169	39.0	282	461.9			SAVASBX	02/17/2025
		2	4.6237	2.8208	39.0	282	462.4			SAVASBX	02/17/2025
		3	4.5989	2.8304	38.5	283	459.9			SAVASBX	02/17/2025
		Avg.	4.6138		39	282	461	50.30	A	SAVASBX	02/17/2025
92M 0103861010	0	1	4.6139	2.8586	38.0	286	461.4			SAVASBX	02/17/2025
		2	4.6353	2.8163	39.2	282	463.5			SAVASBX	02/17/2025
		3	4.5956	2.8462	38.1	285	459.6			SAVASBX	02/17/2025
		Avg.	4.6149		38	284	462	50.30	A	SAVASBX	02/17/2025
92M 0103861011	0	1	4.6319	2.8515	38.4	285	463.2			SAVASBX	02/17/2025
		2	4.6296	2.8527	38.4	285	463.0			SAVASBX	02/17/2025
		3	4.6469	2.8591	38.5	286	464.7			SAVASBX	02/17/2025

ACCEPTANCE TESTING DATA

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Spool	Re - test	Rep	Init - Weight	Final Weight	RC	AWT	PAW	Width	Stat.	Tech	Date
=====											
			Avg. 4.6361		38	286	464	50.30	A	SAVASBX	02/17/2025
=====											
92M 0103861012	0	1	4.6218	2.8400	38.6	284	462.2			SAVASBX	02/17/2025
		2	4.6235	2.8633	38.1	286	462.4			SAVASBX	02/17/2025
		3	4.6185	2.8713	37.8	287	461.8			SAVASBX	02/17/2025
			Avg. 4.6213		38	286	462	50.30	A	SAVASBX	02/17/2025
=====											
92M 0103861013	0	1	4.6563	2.8250	39.3	282	465.6			GILESKX	02/17/2025
		2	4.6336	2.8333	38.9	283	463.4			GILESKX	02/17/2025
		3	4.6234	2.8245	38.9	282	462.3			GILESKX	02/17/2025
			Avg. 4.6378		39	283	464	50.30	A	GILESKX	02/17/2025
=====											
92M 0103861014	0	1	4.5943	2.8144	38.7	281	459.4			GILESKX	02/17/2025
		2	4.6168	2.8187	38.9	282	461.7			GILESKX	02/17/2025
		3	4.6197	2.8251	38.8	282	462.0			GILESKX	02/17/2025
			Avg. 4.6103		39	282	461	50.30	A	GILESKX	02/17/2025
=====											
92M 0103861015	0	1	4.5776	2.8313	38.1	283	457.8			GILESKX	02/17/2025
		2	4.5685	2.8617	37.4	286	456.8			GILESKX	02/17/2025
		3	4.5663	2.8334	37.9	283	456.6			GILESKX	02/17/2025
			Avg. 4.5708		38	284	457	50.30	A	GILESKX	02/17/2025
=====											
92M 0103861016	0	1	4.5650	2.8216	38.2	282	456.5			GILESKX	02/17/2025
		2	4.5423	2.8127	38.1	281	454.2			GILESKX	02/17/2025
		3	4.5474	2.8197	38.0	282	454.7			GILESKX	02/17/2025
			Avg. 4.5516		38	282	455	50.30	A	GILESKX	02/17/2025
=====											
92M 0103861017	0	1	4.6121	2.8471	38.3	285	461.2			GILESKX	02/17/2025
		2	4.5866	2.8644	37.5	286	458.7			GILESKX	02/17/2025
		3	4.6217	2.8445	38.5	284	462.2			GILESKX	02/17/2025
			Avg. 4.6068		38	285	461	50.28	A	GILESKX	02/17/2025
=====											
92M 0103861018	0	1	4.5687	2.8375	37.9	284	456.9			GILESKX	02/17/2025
		2	4.6261	2.8434	38.5	284	462.6			GILESKX	02/17/2025
		3	4.6215	2.8413	38.5	284	462.2			GILESKX	02/17/2025
			Avg. 4.6054		38	284	461	50.28	A	GILESKX	02/17/2025
=====											

IN PROCESS CONTROL

ACCEPTANCE TESTING DATA

Lot Averages Report for Lot: 92M0103861

03/26/2025

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Spool	Retest:	RC	AWT	PAW	Stamp
92M0103861001	0	39	278	458	A
92M0103861002	0	39	280	462	A
92M0103861003	0	39	278	457	A
92M0103861004	0	39	279	457	A
92M0103861005	0	39	280	460	A
92M0103861006	0	39	278	457	A
92M0103861007	0	39	280	463	A
92M0103861008	0	39	282	462	A
92M0103861009	0	39	282	461	A
92M0103861010	0	38	284	462	A
92M0103861011	0	38	286	464	A
92M0103861012	0	38	286	462	A
92M0103861013	0	39	283	464	A
92M0103861014	0	39	282	461	A
92M0103861015	0	38	284	457	A
92M0103861016	0	38	282	455	A
92M0103861017	0	38	285	461	A
92M0103861018	0	38	284	461	A
Averages:		38.7	281.9	460.0	



Product Certification

Page 1 of 2

COA Number:	SEG-COA-27639	Prod Order:	DOM:	Description:
Version:	1	31W0026323	21-Nov-2024	F3C433(FBL2)-50"
Issue Date:	13-Jan-2025	Weave Pattern:	5 Harness Satin	

Specifications

F3C433(FBL2) - GREIGE - 50Inch - HRMS 111546 Rev. 5 Warp Yarn HTA40 E13 3K 200tex Fill Yarn HTA40 E13 3K 200tex

F3C433(FBL2) - GREIGE - 50Inch - MSRR4040 Issue 4 Warp Yarn HTA40 E13 3K 200tex Fill Yarn HTA40 E13 3K 200tex

F3C433(FBL2) - GREIGE - 50Inch - MSRR9992 Issue 12 Warp Yarn HTA40 E13 3K 200tex Fill Yarn HTA40 E13 3K 200tex

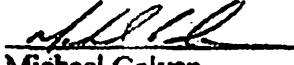
Rolls Tested

Fabric Testing Results

Serial#:	Prod Order:	Property:	Specification:	Result:
31G0000355911	31W0026323	Width	49.5 - 50.5 in	50.2
		Ends per Inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per Inch	17.0 - 18.5 Yarns Per Inch	17.5
		Dried FAW Avg	278 - 292 g/m ²	282
31G0000355912		Width	49.5 - 50.5 in	50.2
		Ends per Inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.5
		Dried FAW Avg	278 - 292 g/m ²	284
31G0000355913		Width	49.5 - 50.5 in	50.2
		Ends per Inch	17.0 - 18.5 Yarns Per Inch	17.7
		Picks per Inch	17.0 - 18.5 Yarns Per Inch	17.6
		Dried FAW Avg	278 - 292 g/m ²	283
31G0000355914		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.5
		Dried FAW Avg	278 - 292 g/m ²	284
31G0000355915		Width	49.5 - 50.5 in	50.2
		Ends per Inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.6

This is to certify that material shipped against this order meets the specification(s) listed above.

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www.hexcel.com


Michael Galvan
Certification Technician



Product Certification

Page 2 of 2

31G0000355915	31W0026323	Dried FAW Avg	278 - 292 g/m ²	283
31G0000355916		Width	49.5 - 50.5 in	50.2
		Ends per Inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per Inch	17.0 - 18.5 Yarns Per Inch	17.5
		Dried FAW Avg	278 - 292 g/m ²	283
31G0000355917		Width	49.5 - 50.5 in	50.2
		Ends per Inch	17.0 - 18.5 Yarns Per Inch	17.7
		Picks per Inch	17.0 - 18.5 Yarns Per Inch	17.5
		Dried FAW Avg	278 - 292 g/m ²	284
31G0000357378		Width	49.5 - 50.5 in	50.2
		Ends per Inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per Inch	17.0 - 18.5 Yarns Per Inch	17.5
		Dried FAW Avg	278 - 292 g/m ²	281
31G0000357379		Width	49.5 - 50.5 in	50.2
		Ends per Inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per Inch	17.0 - 18.5 Yarns Per Inch	17.5
		Dried FAW Avg	278 - 292 g/m ²	284

Certification Exceptions/Additional Information

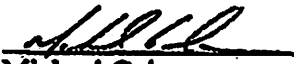
QMS07045, REV. 1, SEC. 5E

WARP: 240220220

FILL: 231220210

This is to certify that material shipped against this order meets the specification(s) listed above.

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Michael Galvan
Certification Technician



TEIJIN CARBON AMERICA, INC.

121 Cardiff Valley Road
Rockwood, TN 37854
Tel. (865) 354-8408
Fax (865) 354-8409

Certificate of Analysis

QAP 004F2 Rev M

Customer: Hexcel Reinforcement Corp (TX) Contact: Patrick Spieckermann; Amy Shehane; Jerry Shehane; Jerry
CustomerPO#: 31PC0013814 Order: SH000919 Email/Fax: Patrick.Spieckermann@hexcel.com; amy.shehane@hexcel.com; jerry.schoeps@; Greg.Early@
Customer P/N: n/a Product: Tenax™-E HTA40 E13 3K 200tex Ship Date: 10/28/2024
Part Number: G305303 Quantity: 672.000kg Certifies To: MS-X-CTC-105 Rev 8; MSRR9317/11
Lot Number: 231220210 Issue 1

We are pleased to advise that the product (s) listed above has been released for shipment. We certify that this lot of Carbon Fiber has been tested by standard test methods and determined to have the following properties:

	<u>Average</u>	<u>CV%</u>
Tensile Strength, MPA (ksi):	4192.7 (608)	3.6
Tensile Modulus, GPa (mil):	238.1 (34.5)	0.8
Elongation, %:	1.68	3.6
Linear Density (without size), g/m:	0.2003	0.7
Size, %:	1.25	6.2
Fiber Density, g/cc:	1.77	
Filament Diameter, µM:	7	
Twist, turn/m:	Never Twisted	
Date of Manufacture:	December 2023	
Manufacturing Location:	Obertruch, Hainaberg, Germany	

Quality Assurance

10/28/2024

These commodities, technology or software are controlled by U. S. Export Administration Regulation. Diversion contrary to U. S. Law is prohibited.

If goods are not received in a reasonable time or if there are any questions, please call 1-800-252-3001 (or (865) 354-8408).
Teijin Carbon America, Inc. 121 Cardiff Valley Road, Rockwood, TN 37854.



TEIJIN CARBON AMERICA, INC.

121 Cardiff Valley Road
Rockwood, TN 37854
Tel. (888) 354-8408
Fax (888) 354-8409

Certificate of Analysis

QAP 00482 Rev M

Customer:	Hexcel Reinforcement Corp (TX)	Contact:	Patrick Speckermann; Amy Shehane; Jerry	Email/Fax:	Patrick.Speckermann@hexcel.com; amy.shehane@hexcel.com; jerry.tschoepe@; Greg.Early@
CustomerPO#:	31PO0013614	Order:	SH000919	Ship Date:	10/28/2024
Customer P/N:	n/a	Product:	Tenax TM -E HTA40 E13 3K 200tex		
Part Number:	G305303	Quantity:	3096.000 kg	Certified To:	MS-X-CTC-105 Rev B; MSRR8317/11 Issue 1
Lot Number:	240220220				

We are pleased to advise that the product (s) listed above has been released for shipment. We certify that this lot of Carbon Fiber has been tested by standard test methods and determined to have the following properties:

	<u>Average</u>	<u>CV%</u>
Tensile Strength, MPA (ksi):	4216.6 (612)	3.7
Tensile Modulus, GPa (msi):	238.4 (34.6)	0.9
Elongation, %:	1.67	3.5
Linear Density (without size), g/m:	0.1998	0.7
Size, %:	1.26	8.1
Fiber Density, g/cc:	1.77	
Filament Diameter, µm:	7	
Twist, turn/m:	Never Twisted	
Date of Manufacture:	February 2024	
Manufacturing Location:	Oberbruch, Hainberg, Germany	

Quality Assurance

10/28/2024

These commodities, technology or software are controlled by U. S. Export Administration Regulation. Diversion contrary to U. S. Law is prohibited.

If goods are not received in a reasonable time or if there are any questions, please call 1-800-282-3001 (or (888) 354-8408).
Teijin Carbon America, Inc. 121 Cardiff Valley Road, Rockwood, TN 37854.

ROLL CONDITION LOG

Batch Number 92M0103861	Serial Number 92M0103861001	Fiber F3C 433	Resin M65	Prepreg Width 50
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Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
6	50	1	0.3	SLUB
136	50	0	0.0	FUZZBALL

Final Footage 225	Net Feet 224	Credit Method A	Defect % 0.037%	Date 2/17/2025
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Inspection Stamp 	Date 2/17/2025
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ROLL CONDITION LOG

Batch Number 92M0103861	Serial Number 92M0103861002	Fiber F3C 433	Resin M65	Prepreg Width 50
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Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
125	50	12	4.2	SPLICE

Final Footage 215	Net Feet 214	Credit Method A	Defect % 0.465%	Date 2/17/2025
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Inspection Stamp 	Date 2/17/2025
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ROLL CONDITION LOG

Batch Number #VALUE!	Serial Number 92M0103861003	Fiber F3C 433	Resin M65	Prepreg Width 50
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Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
22	50	0	0.0	FUZZBALL
40	50	0	0.0	FUZZBALL
119	50	0	0.0	FUZZBALL
197	50	0	0.0	FUZZBALL

Final Footage 232	Net Feet 232	Credit Method A	Defect % 0.000%	Date 2/17/2025
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Inspection Stamp 	Date 2/17/2025
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ROLL CONDITION LOG

Batch Number	Serial Number	Fiber	Resin	Prepreg Width
92M0103861	92M0103861004	F3C 433	M65	50

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
78	50	0	0.0	FUZZBALL
90	50	0	0.0	FUZZBALL
107	50	24	8.3	SPLICE
115	50	24	8.3	SPLICE
130	50	12	4.2	RESIN LEAN

Final Footage	Net Feet	Credit Method	Defect %	Date
215	210	A	2.326%	2/17/2025

Inspection Stamp	Date
	2/17/2025

ROLL CONDITION LOG

Batch Number 92M0103861	Serial Number 92M0103861005	Fiber F3C 433	Resin M65	Prepreg Width 50
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Feet From Start	Defect Width (Inches)	Defect Length (Inches)	Credit Amount (sq. ft.)	Defect Type Remarks
110	50	0	0.0	FUZZBALL

Final Footage 225	Net Feet 225	Credit Method A	Defect % 0.000%	Date 2/17/2025
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Inspection Stamp 	Date 2/17/2025
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ROLL CONDITION LOG

Batch Number	Serial Number	Fiber	Resin	Prepreg Width
92M0103861	92M0103861006	F3C 433	M65	50

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
25	50	12	4.2	FLOAT
90	50	0	0.0	FUZZBALL
129	50	1	0.3	FLOAT
145	50	0	0.0	FUZZBALL
195	50	24	8.3	CURED RESIN
210	50	12	4.2	RESIN LEAN

Final Footage	Net Feet	Credit Method	Defect %	Date
220	215	A	1.856%	2/17/2025

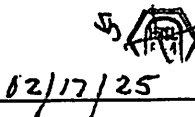
Inspection Stamp	Date
	2/17/2025

ROLL CONDITION LOG

Batch Number 92M0103861	Serial Number 92M0103861007	Fiber F3C 433	Resin M65	Prepreg Width 50
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Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
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Final Footage 225	Net Feet 225	Credit Method A	Defect % 0.000%	Date 2/17/2025
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Inspection Stamp  	Date 2/17/2025
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ROLL CONDITION LOG

Batch Number 92M0103861	Serial Number 92M0103861008	Fiber F3C 433	Resin M65	Prepreg Width 50
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Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
53	50	1	0.3	FLOAT
127	50	1	0.3	FUZZBALL

Final Footage 221	Net Feet 220	Credit Method A	Defect % 0.075%	Date 2/17/2025
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Inspection Stamp 	Date 2/17/2025
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ROLL CONDITION LOG

Batch Number 92M0103861	Serial Number 92M0103861009	Fiber F3C 433	Resin M65	Prepreg Width 50
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Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
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Final Footage 229	Net Feet 229	Credit Method A	Defect % 0.000%	Date 2/17/2025
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Inspection Stamp 	Date 2/17/2025
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ROLL CONDITION LOG

Batch Number	Serial Number	Fiber	Resin	Prepreg Width
92M0103861	92M0103861010	F3C 433	M65	50

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
4	50	24	8.3	CURED RESIN

Final Footage	Net Feet	Credit Method	Defect %	Date
212	210	A	0.943%	2/17/2025

Inspection Stamp	Date
	2/17/2025

ROLL CONDITION LOG

Batch Number 92M0103861	Serial Number 92M0103861011	Fiber F3C 433	Resin M65	Prepreg Width 50
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Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
91	50	3	1.0	MISPICK

Final Footage 239	Net Feet 238	Credit Method A	Defect % 0.105%	Date 2/17/2025
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Inspection Stamp 	Date 2/17/2025
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ROLL CONDITION LOG

Batch Number 92M0103861	Serial Number 92M0103861012	Fiber F3C 433	Resin M65	Prepreg Width 50
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Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
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Final Footage 192	Net Feet 192	Credit Method A	Defect % 0.000%	Date 2/17/2025
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Inspection Stamp 	Date 2/17/2025
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ROLL CONDITION LOG

Batch Number 92M0103861	Serial Number 92M0103861013	Fiber F3C 433	Resin M65	Prepreg Width 50
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Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
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Final Footage 230	Net Feet 230	Credit Method A	Defect % 0.000%	Date 2/17/2025
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Inspection Stamp 	Date 2/17/2025
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ROLL CONDITION LOG

Batch Number 92M0103861	Serial Number 92M0103861014	Fiber F3C 433	Resin M65	Prepreg Width 50
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Feet From Start	Defect Width (Inches)	Defect Length (Inches)	Credit Amount (sq. ft.)	Defect Type Remarks
85	50	.1	0.3	SLUB

Final Footage 216	Net Feet 215	Credit Method A	Defect % 0.039%	Date 2/17/2025
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Inspection Stamp 	Date 2/17/2025
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ROLL CONDITION LOG

Batch Number 92M0103861	Serial Number 92M0103861015	Fiber F3C 433	Resin M65	Prepreg Width 50
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Feet From Start	Defect Width (Inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
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Final Footage 230	Net Feet 230	Credit Method A	Defect % 0.000%	Date 2/18/2025
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Inspection Stamp 	Date 2/18/2025
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ROLL CONDITION LOG

Batch Number 92M0103861	Serial Number 92M0103861016	Fiber F3C 433	Resin M65	Prepreg Width 50
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Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
37	50	0	0.0	FUZZBALL

Final Footage 208	Net Feet 208	Credit Method A	Defect % 0.000%	Date 2/18/2025
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Inspection Stamp 	Date 2/18/2025
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ROLL CONDITION LOG

Batch Number 92M0103861	Serial Number 92M0103861017	Fiber F3C 433	Resin M65	Prepreg Width 50
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Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
70	50	1	0.3	SLUB
120	50	180	62.5	RESIN LEAN

Final Footage 230	Net Feet 214	Credit Method A	Defect % 6.558%	Date 2/18/2025
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Inspection Stamp 	Date 2/18/2025
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ROLL CONDITION LOG

Batch Number 92M0103861	Serial Number 92M0103861018	Fiber F3C 433	Resin M65	Prepreg Width 50
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Feet From Start	Defect Width (Inches)	Defect Length (Inches)	Credit Amount (sq. ft.)	Defect Type Remarks
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Final Footage 220	Net Feet 220	Credit Method A	Defect % 0.000%	Date 2/18/2025
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Inspection Stamp 	Date 2/18/2025
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